

Rohit Gupta

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Education

2019 - 2025	PhD. in Computer Science , University of Central Florida, Orlando
2015 - 2017	M.Tech. in Computer Science and Engineering , IIT Kanpur, Kanpur, India
2010 - 2014	B.Tech. in Electrical Engineering , IIT Kanpur, Kanpur, India

Selected Publications

ACL 2026 Oral (Top 5%)	ViLLE: Video LLM Embeddings for Retrieval R Gupta , J Unnikrishnan, F Fei, S Liu, S Tran, M Shah <i>Pre-Print: https://arxiv.org/abs/2604.12148</i>
CVPR 2026 Highlight	VRR-QA: Visual Relational Reasoning in Videos Beyond Explicit Cues Sirnam S, R Gupta , P Kulkarni, D Shatwell, J Santiago, N Siddiqui, J Fioresi, M Shah <i>Link: https://openaccess.thecvf.com/content/CVPR2026/html/Swetha_VRR-QA</i>
CVPR 2026	VidTAG: Temporally Aligned Video to GPS Geolocalization with Denoising Sequence Prediction at a Global Scale PP Kulkarni, R Gupta , M Shah <i>Link: https://openaccess.thecvf.com/content/CVPR2026/html/Kulkarni_VidTAG</i>
ECCV 2026	The Telephone Game: Evaluating Semantic Drift in Unified Models. S Mollah, R Gupta , Sirnam S, Q Liu, A Munir, M Shah <i>arXiv: https://arxiv.org/abs/2509.04438</i>
ECCV 2026	SB-Bench: Stereotype Bias Benchmark for Large Multimodal Models. V Narnaware, A Vayani, R Gupta , Sirnam S, M Shah <i>arXiv: https://arxiv.org/abs/2502.08779</i>
ECCV 2024 Citations: 11	Open Vocabulary Multi-Label Video Classification R Gupta , MN Rizve, A Tawari, J Unnikrishnan, S Tran, M Shah <i>(Link: https://www.ecva.net/papers/eccv_2024/papers_ECCV/papers/05599.pdf)</i>
CVPR 2023 Citations: 41	Class Prototypes based Contrastive Learning for Classifying Multi-Label and Fine-Grained Educational Videos R Gupta , A Roy, S Kim, C Christensen, T Grindal, S Gerard, M Cincebeaux, A Divakaran, M Shah <i>Link: https://openaccess.thecvf.com/content/CVPR2023/html/Gupta_Class_Prototypes_...</i>
AAAI 2023 Citations: 15	Contrastive Self-Supervised Learning Leads to Higher Adversarial Susceptibility R Gupta , N Akhtar, A Mian, M Shah <i>Link: https://ojs.aaai.org/index.php/AAAI/article/view/26733</i>
WACV 2026 Citations: 16	GAEA: A Geolocation Aware Conversational Assistant R Campos, A Vayani, PP Kulkarni, R Gupta , A Dutta M Shah <i>Link: https://openaccess.thecvf.com/content/WACV2026/html/Campos_GAEA</i>
ICPR 2020 Citations: 152	RescueNet: Joint building segmentation and damage assessment from satellite imagery R Gupta , M Shah <i>Link: https://ieeexplore.ieee.org/document/9412295</i>
CVIU, Jun '22 Citations: 276	TCLR: Temporal Contrastive Learning for Video Representation I Dave, R Gupta , M N Rizve, M Shah <i>Link: https://www.sciencedirect.com/science/article/pii/S1077314222000376</i>
MediaEval 2018 Citations: 18	Linear Models for Video Memorability Prediction Using Visual and Semantic Features R Gupta , K Motwani <i>Link: http://ceur-ws.org/Vol-2283/MediaEval_18_paper_31.pdf</i>

Work Experience

Applied Scientist II (L5), Amazon Prime Video Science

August 2025-Present

- Developed a fast broad catalog video clips search engine and a highly adaptive state of the art within video clip search LLM agent.
- Search work is being leveraged by Amazon studios editors resulting in \$20 million worth of projected savings.
- Contributed to a hybrid audio-visual model for vending suitable cuepoints for Ads, increasing ad inventory by \$70 million annually.
- Built eval and training datasets for a text embedding model specialized for semantic matching of movie and TV show titles.
- Carried out embedding model training on 1 million+ titles, achieving upto 15% improvement in recommendation tasks.

Applied Scientist Intern, Amazon Nile, Rufus Multi-Modal Team

May-Nov 2024

- Developed a single model capable of solving a wide spectrum of video understanding tasks: Question Answering, Video Retrieval, Localization etc
- Built an unified Video LLM with the capabilities of outputting text as well as embeddings in order to solve all the tasks.
- Carried out medium scale distributed post-training on 10 million+ videos.

Applied Scientist Intern, Amazon Search Science and AI, M5 Team

May-Nov 2023, Palo Alto

- Worked on multi-label open vocabulary video classification; recognizing objects and actions in videos not present in training data.
- Work resulted in a successful publication at ECCV 2024 main conference.

Research Intern, SRI International

May- Aug 2022, Menlo Park

- Was tasked with solving the problem of classifying youtube education videos into fine-grained topic categories within subjects.
- Work resulted in a successful poster publication at CVPR 2023 main conference, and was invited as an Oral presentation at a workshop.

Achievements

2025,2026	Organizer , Workshop on Video Large Language Models, CVPR
2022-present	Reviewer , CVPR, ECCV, ICCV, AAAI, ICLR, NeurIPS, WACV, IEEE (TPAMI, TIP, TNNLS, TCSVT)
2023	Oral Presentation , Workshop on Large Scale Holistic Video Understanding, CVPR
2021	1st Place and Jury Prize , VI-Priors Action Recognition Challenge, ICCV
2019	Fellowship , ORCGS Doctoral Fellowship, UCF
2010	National Rank 433 (Top 0.1%) , Joint Entrance Exam, Indian Institutes of Technology